5 Normal Multivariate Distribution

MA2003B Application of Multivariate Methods in Data Science

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Definition and Basic Properties

Definition

Let μ be a (row) vector and let Σ be a symmetric positive definite matrix. We say that a random (row) vector X has **normal multivariate distribution** with mean μ and covariance matrix Σ, in symbols $X \sim \mathscr{N}_p(\mu, \Sigma)$, if its probability density is given by

$$\frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(X-\mu)\Sigma^{-1}(X-\mu)^T\right) dX$$

Remark

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is the **generalized variance** of the distribution. It can be interpreted as a measure of the "spread" of the distribution in the p-dimensional space.
4. Notice that the term $(X-\mu)\Sigma^{-1}(X-\mu)^T$ is the square of the Mahalanobis distance between the random vector X and the mean μ.

Just like in the univariate case, the normal multivariate distribution is singled out among all other possible distributions by the Central Limit Theorem, that roughly states that the sum of a large number of independent and identically distributed random variables (properly standardized) converges to a normal distribution.

Multivariate Central Limit Theorem

Theorem

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of independent and identically distributed random vectors in $\mathbb{R}^p$ with mean vector μ and covariance matrix Σ. Define the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Then, as $n \to \infty$,

$$\sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}_p(0, \Sigma),$$

where $\xrightarrow{d}$ denotes convergence in distribution and $\mathcal{N}_p(0,\Sigma)$ is the p-dimensional multivariate normal distribution with mean vector 0 and covariance matrix Σ.

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Furthermore, important statistical methods, like linear regression, principal component analysis and clustering do not yield good results when applied to a normal multivariate distribution.

Normal Multivariate Contours

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The probability distribution of these contours is controlled by the χ^2 distribution in the following sense:

Let $0 \leq \varpi \leq 1$ be a probability level, and let c be such that

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Then, the contour that contains a proportion ϖ of the probability mass of the distribution is given by the set of points X that satisfy the equation

$$(X - \mu)\Sigma^{-1}(X - \mu)^T = c$$

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The color gradient represents the density of the distribution, with darker colors indicating higher density.

Contours of the Bivariate Normal

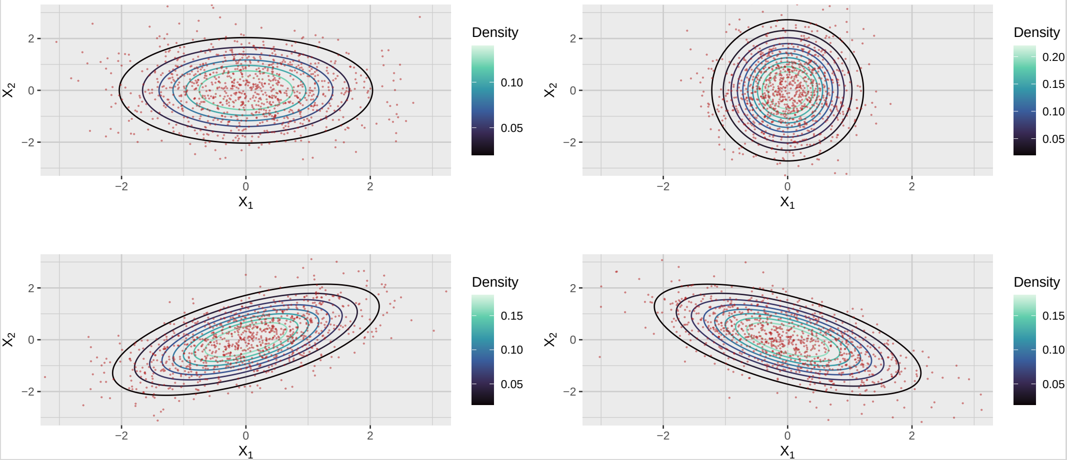

Figure 1: Bivariate Normal Distribution with Different Covariance Matrices

Standardization of the Normal Multivariate Distribution

Just like in the univariate case, given a random vector $X \sim \mathcal{N}_p(\mu, \Sigma)$, we can standardize it to obtain a new random vector Z that follows the standard multivariate normal distribution $\mathcal{N}_p(0, I_p)$, where I_p is the identity matrix of size p.

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To describe this standardization process, we first need to remember some linear algebra results.

Singular Value Decomposition

Theorem

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$$\Sigma = Q \Lambda Q^T$$

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Remark

Remember that a matrix Q is orthogonal if $Q^T Q = I_p$, where I_p is the identity matrix of size p. This means that the columns of Q are orthonormal vectors, and thus they form a basis for $\mathbb{R}^p$.

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In the case of the covariance matrix Σ, the SVD provides a way to express it in terms of its **singular values** (the diagonal entries of Λ, also known as its **eigenvalues**) and its **singular vectors** (the columns of Q, also known as its **eigenvectors**).

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This decomposition is particularly useful for understanding the geometry of the multivariate normal distribution.

Notice that, since Σ is positive definite, the diagonal entries of

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_p \end{bmatrix}$$

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are all positive.

In particular, we can define a new matrix

$$D = \begin{bmatrix} \sqrt{\lambda_1} & 0 & \cdots & 0 \\ 0 & \sqrt{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{\lambda_p} \end{bmatrix}$$

satisfying $D^2 = \Lambda$.

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In particular, we can define the **standardization** of the random vector X as

$$Z = (X - \mu)\sigma^{-1} = (X - \mu)QD^{-1} = (X - \mu)Q\Lambda^{-1/2}$$

in which case $Z \sim \mathcal{N}_p(0, I_p)$.

Closure under Linear Transformations

Theorem

Let $X \sim \mathcal{N}_p(\mu, \Sigma)$ be a random vector with normal multivariate distribution. Then, for any matrix A of size $m \times p$ and any vector b of size m, the random vector

$$Y = AX + b$$

also has a normal multivariate distribution, specifically $Y \sim \mathcal{N}_m(A\mu + b, A\Sigma A^T)$.

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In practice, this theorem is often used to derive the distribution of linear combinations of random variables, which is a common operation in multivariate statistics.

Example

Let $X \sim \mathcal{N}_2(\mu, \Sigma)$ be a random vector with mean $\mu = (1, 2)$ and covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 2 \end{bmatrix}$$

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In other words,

$$Y_1 = 2X_1 + X_2$$
$$Y_2 = 3X_2 + 1$$

Example

Using the closure under linear transformations theorem, we can find the mean and covariance of Y:

$$\text{Mean: } A\mu + b = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

$$\text{Covariance: } A\Sigma A^T = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0.5 \\ 0.5 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 9 \\ 9 & 18 \end{bmatrix}$$

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Thus,

$$Y \sim \mathcal{N}_2\left(\begin{bmatrix} 4 \\ 7 \end{bmatrix}, \begin{bmatrix} 8 & 9 \\ 9 & 18 \end{bmatrix} \right)$$

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This is a very useful property, as it allows us to analyze subsets of variables while still maintaining the normality assumption.

Multivariate Normality Testing

There are several statistical tests to check whether a given dataset follows a multivariate normal distribution. These tests are essential in multivariate statistics, as many methods assume normality. Here are some of the most commonly used tests, together with their implementation in R:

1. **Mardia's Test:** Based on multivariate skewness and kurtosis. For a multivariate normal distribution, skewness should be close to 0 and kurtosis close to $p(p+2)$, where p is the number of variables. Implemented in R via `MVN::mvn(your_data, mvn_test = "mardia")`.

2. **Henze-Zirkler (HZ) Test:** Uses the empirical characteristic function to measure deviations from normality. It has good global power and is suitable for moderate dimensions. Implemented in R via `MVN::mvn(your_data, mvn_test = "hz")`.

3. **Royston's Test:** A multivariate extension of the Shapiro-Wilk test, based on the product of univariate Shapiro-Wilk statistics applied to principal components. Easy to interpret, though power decreases in high dimensions. Implemented in R via `MVN::mvn(your_data, mvn_test = "royston")`.
4. **Doornik-Hansen Test:** Combines skewness and kurtosis into a single omnibus test, providing a robust alternative to Mardia. Implemented in R via `MVN::mvn(your_data, mvn_test = "dh")`.
5. **Graphical Methods (Chi-square Q-Q plots):** Plot the Mahalanobis distances of the data against the quantiles of a chi-square distribution. Deviations from a straight line indicate departures from multivariate normality. Implemented in R via `multivariate_diagnostic_plot(data = your_data, type = "qq")` .